

Patterns of Hospital Admissions and Healthcare Utilisation: A Descriptive and Inferential Analysis of a Synthetic Electronic Health Record Dataset

Submitted in Partial Fulfilment of Academic Research Requirements

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2026

ABSTRACT

This study presents a descriptive and inferential statistical analysis of a synthetic hospital admissions dataset comprising 10,000 patient records and 15 clinical, demographic, and financial variables. The dataset, modelled after real-world Electronic Health Records (EHR), was examined to identify patterns in disease prevalence, demographic distribution, financial impact, and the determinants of hospital billing. The analysis reveals that chronic non-communicable diseases—namely Arthritis, Cancer, and Hypertension—constitute the most prevalent conditions among admitted patients, with the 60–79 age group accounting for the highest volume of admissions. Financial analysis indicates that Cancer and Diabetes incur comparable average billing amounts of approximately USD 25,300, while inferential tests demonstrate that neither patient age ($r = 0.003$), admission type (ANOVA $p = 0.970$), nor gender (Chi-Square $p = 0.958$) significantly predicts billing cost or diagnosis outcome. These findings suggest that treatment complexity and condition type are the primary drivers of healthcare expenditure, and that resource allocation should be strategically directed towards adult and geriatric chronic disease management. The implications of these results are discussed in the context of healthcare planning, insurance policy, and hospital administration.

Keywords: hospital admissions, chronic non-communicable diseases, healthcare billing, EHR analysis, descriptive statistics, inferential statistics

1. INTRODUCTION

The efficient management of healthcare systems depends critically on the ability of administrators, clinicians, and policymakers to understand the demographic, clinical, and financial profiles of patient populations. In both high-income and developing-country contexts, hospital admission data represents one of the most information-rich sources for understanding disease burden, resource utilisation, and financial sustainability within health facilities. The systematic analysis of Electronic Health Records (EHR) has emerged as a powerful methodology for extracting actionable insights from large volumes of patient data, enabling evidence-based decision-making at the institutional and national levels.

This paper presents a comprehensive analysis of a synthetic hospital admissions dataset containing 10,000 patient records drawn from a Kaggle-hosted repository designed to mirror the structure of real-world EHR systems. The dataset captures a patient's journey from admission through discharge and encompasses demographic variables (age, gender, blood type), clinical variables (medical condition, medication, test results, admission type), and financial variables (billing amount, insurance provider). Although the dataset is synthetic in origin, its structure closely replicates the format of data that would be collected through hospital management systems or manual intake registration in clinical settings such as those found in Mogadishu, Somalia.

The study is guided by two overarching research objectives: first, to describe the distribution of medical conditions, demographic characteristics, and financial costs within the dataset; and second, to test a set of inferential hypotheses regarding the relationships between key variables, including the association between patient age and billing amount, the effect of admission type on cost, and the independence of gender from medical diagnosis. By addressing these objectives, the study aims to generate findings of practical relevance to hospital boards, public health planners, and insurance providers.

2. BACKGROUND AND LITERATURE CONTEXT

The global burden of non-communicable diseases (NCDs) has been extensively documented, with conditions such as cancer, diabetes, hypertension, and arthritis collectively accounting for the majority of hospital admissions and healthcare expenditure in countries across income levels (World Health Organisation, 2023). In sub-Saharan Africa, the epidemiological transition from predominantly infectious to chronic disease burdens has placed increasing strain on health systems that were historically oriented towards communicable disease management. Somalia, in particular, faces compounding challenges of fragile health infrastructure, limited insurance penetration, and a growing prevalence of NCDs, making data-driven hospital planning an urgent priority.

Research in health informatics has consistently demonstrated the value of EHR-based analyses for identifying high-risk patient populations and optimising resource allocation. Studies have shown that age remains one of the most significant predictors of hospital admission frequency, with elderly populations typically exhibiting higher rates of complex comorbidities and longer lengths of stay (Barnett et al., 2012). However, the relationship between patient age and billing amount is less straightforward, as cost is often more closely tied to diagnosis complexity, treatment protocol, and hospital pricing structures than to demographic characteristics alone.

The role of insurance in moderating healthcare financial burden has also received considerable scholarly attention. Uninsured or underinsured patients frequently incur higher out-of-pocket costs and may defer seeking care, ultimately presenting with more advanced disease stages and incurring greater treatment costs. The analysis of insurance provider data within hospital admission records therefore offers a window into equity of access and financial protection within health systems.

This study contributes to the existing literature by applying both descriptive and inferential analytical methods to a structured EHR-type dataset, with findings interpretable in the context of healthcare planning in low-to-middle-income settings. Given the absence of locally validated Somali hospital datasets, the use of a synthetic dataset that mirrors EHR structure provides a methodologically sound proxy for developing analytical frameworks applicable to real-world data.

3. METHODOLOGY

3.1 Dataset Description

The dataset analysed in this study is a synthetic healthcare record comprising 10,000 patient entries and 15 variables. The dataset was sourced from the Kaggle open data repository and was designed to replicate the structure and complexity of real-world Electronic Health Records. It contains intentional data quality issues, including inconsistencies in text casing and the presence of outliers, to simulate the imperfections commonly encountered in operational clinical databases. The variables encompass three domains: demographic (Name, Age, Gender, Blood Type), clinical (Medical Condition, Doctor, Hospital, Admission Type, Medication, Test Results, Date of Admission, Discharge Date, Room Number), and financial (Billing Amount, Insurance Provider).

3.2 Research Questions

The study was structured around the following six research questions, organised into descriptive and inferential categories:

Descriptive questions: (1) What are the three most prevalent medical conditions in the dataset? (2) Which age group accounts for the highest number of hospital admissions? (3) What is the average billing amount for Cancer patients compared to Diabetes patients?

Inferential questions: (4) Is there a significant correlation between patient age and billing amount? (5) Does admission type significantly affect billing amount? (6) Is there a statistically significant relationship between gender and medical condition?

3.3 Analytical Methods

Descriptive analysis was conducted using frequency counts, percentage distributions, and grouped summary statistics including means. Disease prevalence was assessed via tabulation of the Medical Condition variable. Demographic analysis was performed by categorising the Age variable into five standardised age brackets: 0–19, 20–39, 40–59, 60–79, and 80–100.

Inferential analyses employed three statistical techniques appropriate to the data types and research questions. Pearson's correlation coefficient was applied to examine the linear relationship between Age (continuous) and Billing Amount (continuous). A one-way Analysis of Variance (ANOVA) was conducted to test for differences in mean Billing Amount across the three levels of Admission

Type (Emergency, Urgent, and Elective). Finally, the Chi-Square Test of Independence was used to assess the association between the categorical variables of Gender and Medical Condition. A significance threshold of $\alpha = 0.05$ was adopted for all inferential tests.

4. RESULTS AND FINDINGS

4.1 Disease Prevalence

The distribution of medical conditions across the 10,000 patient records is presented in Table 1. The analysis reveals a remarkably narrow distribution across the six recorded diagnoses, with each condition accounting for between 16.3% and 16.9% of total admissions. Arthritis and Cancer were tied as the most prevalent conditions, each recorded in 1,687 cases, followed by Hypertension (1,676), Diabetes (1,659), Asthma (1,658), and Obesity (1,633).

Medical Condition	Count	Percentage (%)
Arthritis	1,687	16.87%
Asthma	1,658	16.58%
Cancer	1,687	16.87%
Diabetes	1,659	16.59%
Hypertension	1,676	16.76%
Obesity	1,633	16.33%
Grand Total	10,000	100.00%

Table 1. Distribution of Medical Conditions Among Hospital Admissions (N = 10,000)

The near-uniform distribution of conditions is a notable feature of this dataset. The absence of a single dominant diagnosis suggests that the facility serves a diverse patient population affected by multiple chronic non-communicable conditions simultaneously, rather than operating as a specialist centre for any particular illness. This pattern has direct implications for procurement, staffing, and formulary management, as the facility must maintain adequate capacity across all six condition categories concurrently.

4.2 Demographic Distribution by Age Group

Table 2 presents the distribution of hospital admissions across five age categories. The 60–79 age group recorded the highest admission volume at 2,942 patients (29.4%), followed by the 40–59 group (2,930; 29.3%) and the 20–39 group (2,908; 29.1%). The 80–100 age bracket accounted for 916 admissions (9.2%), while the 0–19 cohort represented the smallest proportion of the patient population, with only 304 cases (3.0%).

Age Group	Number of Admissions	Percentage (%)
0–19	304	3.04%
20–39	2,908	29.08%
40–59	2,930	29.30%
60–79	2,942	29.42%
80–100	916	9.16%
Grand Total	10,000	100.00%

Table 2. Distribution of Hospital Admissions by Age Group (N = 10,000)

Collectively, patients aged 20 to 79 years accounted for 87.8% of all admissions, underscoring the predominantly adult and middle-aged character of the patient population. The relative under-representation of paediatric patients (0–19 years at 3%) has significant implications for resource allocation, suggesting that the facility's strategic investment priorities should be oriented towards adult internal medicine and geriatric care rather than paediatric services.

4.3 Financial Analysis: Cancer Versus Diabetes Billing

The average billing amounts for the two conditions of primary financial interest are presented in Table 3. The mean billing amount for Cancer patients was USD 25,348.84, compared to USD 25,197.53 for Diabetes patients—a difference of approximately USD 151.31.

Medical Condition	Average Billing Amount (USD)
Cancer	\$25,348.84
Diabetes	\$25,197.53

Table 3. Average Billing Amount for Cancer and Diabetes Patients

Although Cancer treatment incurs marginally higher costs, the similarity in average billing amounts across both conditions is the more analytically significant observation. Both figures indicate that the management of these chronic conditions imposes a substantial financial burden on patients and payers alike. The high average cost of treatment in both categories underscores the critical importance of comprehensive insurance coverage and hospital-level financial planning for sustainable chronic disease management.

4.4 Correlation Between Age and Billing Amount

A Pearson correlation analysis was conducted to examine the relationship between patient age and hospital billing amount. The resulting coefficient ($r = 0.00344$) indicates an extremely weak positive association that approaches zero, effectively demonstrating statistical independence between the two variables. This finding is presented in Table 4.

Variable	Age	Billing Amount
Age	1.000	—
Billing Amount	0.003	1.000

Table 4. Pearson Correlation Matrix: Age and Billing Amount

The near-zero correlation coefficient indicates that a patient's age provides negligible predictive value for the amount they will be charged during a hospital admission. The billing amount appears to fluctuate independently of patient age, suggesting that other factors—most likely the specific nature of the medical diagnosis and the complexity or duration of treatment—are the primary determinants of cost.

4.5 Effect of Admission Type on Billing Amount (ANOVA)

A one-way ANOVA was performed to test whether the three categories of Admission Type (Emergency, Urgent, and Elective) produce statistically different mean billing amounts. The test yielded an F-statistic with an associated P-value of 0.9701, far exceeding the 0.05 significance threshold. The mean billing amounts across all three admission categories were consistently close to USD 25,600. Accordingly, the null hypothesis of equal means was retained, and no statistically significant difference was observed.

This result implies that, regardless of the urgency or planned nature of a patient's admission, the financial cost imposed by the hospital remains essentially uniform. This may indicate that the hospital applies a standardised pricing model based on diagnosis-related groups or treatment protocols rather than varying charges based on admission circumstances. The practical implication is that cost containment strategies must focus on clinical variables rather than the mode of admission.

4.6 Relationship Between Gender and Medical Condition (Chi-Square Test)

The Chi-Square Test of Independence was applied to examine whether a statistically significant relationship exists between patient gender and medical condition. The test produced a Chi-Square statistic of 0.6974 and a P-value of 0.9576. As the P-value substantially exceeds the $\alpha = 0.05$ threshold, the null hypothesis of independence was retained.

This finding indicates that the medical conditions recorded in this dataset are distributed independently of patient gender. The diagnoses of Cancer, Diabetes, Arthritis, Hypertension, Asthma, and Obesity appear to affect male and female patients in statistically equivalent proportions. Consequently, gender does not emerge as a differentiating factor for disease risk in this dataset, and gender-specific targeted interventions would not be justified by these findings alone.

5. DISCUSSION

The collective findings of this study generate a coherent and internally consistent picture of a hospital that primarily serves an adult and aging population burdened by a diverse range of chronic non-communicable diseases, at a relatively uniform and substantial financial cost.

The near-identical prevalence rates across the six recorded medical conditions—each hovering between 16.3% and 16.9%—suggest that no single disease dominates the admission profile. While this may partly reflect the synthetic and evenly distributed nature of the dataset, it also highlights the importance of avoiding a narrow disease-specific focus in hospital planning. Facilities that concentrate disproportionate resources on a single condition risk inadequate capacity for equally prevalent conditions. The balanced prevalence observed here would ideally call for an integrated

chronic disease management programme capable of addressing all six conditions in a coordinated manner.

The demographic findings are consistent with global trends documenting an aging patient population in hospital settings. The concentration of admissions in the 40–79 age range aligns with the known peak incidence period for most NCDs, including hypertension, arthritis, and cancer. The low volume of paediatric admissions (3%) may reflect lower disease severity or differential care-seeking behaviour in younger populations, or may simply indicate that this facility serves primarily as an adult care centre. Regardless, the data clearly supports a strategic focus on adult and geriatric medicine.

The financial analyses produced two notable results. First, the near-parity in billing amounts between Cancer and Diabetes patients challenges any assumption that cancer care is dramatically more expensive than the management of metabolic conditions such as diabetes in this particular institutional context. Both conditions generate average bills of approximately USD 25,300, reflecting the sustained and resource-intensive nature of chronic disease management more broadly. Second, the lack of significant variation in billing by admission type or patient age suggests that the hospital's pricing structure is condition-driven rather than circumstance-driven, a finding with important implications for insurance product design and premium calculation.

The inferential findings collectively indicate that demographic factors—age and gender—are poor predictors of either financial cost or disease diagnosis in this dataset. This shifts analytical attention toward clinical and treatment-related variables as the more meaningful explanatory factors. Future analyses would benefit from examining the relationship between length of hospital stay, medication type, test results, and billing amount, which were not the focus of the present study.

6. CONCLUSION

This study conducted a systematic descriptive and inferential analysis of a 10,000-record synthetic hospital admissions dataset modelled on Electronic Health Record structures. The key findings can be summarised as follows: Arthritis and Cancer were the most prevalent conditions, each affecting 16.87% of the patient population; the 60–79 age group recorded the highest admission volume; Cancer and Diabetes imposed comparable financial burdens averaging approximately USD 25,300

per patient; and none of the tested demographic variables—age, admission type, or gender—demonstrated a statistically significant relationship with billing amount or medical diagnosis.

Taken together, these findings reinforce the understanding that chronic non-communicable disease management is the central challenge and cost driver for the type of facility represented in this dataset. Demographic variables, while important for understanding patient profiles, do not appear to be significant determinants of the financial or clinical outcomes examined. Strategic decisions regarding budgeting, staffing, and service planning should therefore be anchored in the clinical complexity of conditions treated rather than in the demographic composition of the patient population.

The methodological framework employed in this study—combining descriptive profiling with correlation, ANOVA, and Chi-Square testing—is transferable to real hospital datasets in low-resource settings, including those in Mogadishu or broader East Africa, where evidence-based health planning remains a critical need.

7. RECOMMENDATIONS

Based on the findings of this study, the following recommendations are presented for consideration by hospital administrators, public health policymakers, and insurance providers:

1. **Invest in Integrated Chronic Disease Programmes:** Given the near-equal prevalence of six distinct NCDs, the facility should establish or strengthen multi-disciplinary chronic disease management programmes rather than disease-specific silos. Shared diagnostic infrastructure, pooled pharmacy procurement, and cross-trained clinical staff would improve efficiency and patient outcomes across all conditions.
2. **Prioritise Geriatric and Adult Care Capacity:** The dominance of the 40–79 age bracket in admission records necessitates deliberate investment in adult and geriatric care capacity, including specialist physicians, appropriate ward infrastructure, and tailored rehabilitation services. Paediatric services, while essential, should be calibrated proportionally to the low observed demand.

3. Strengthen Insurance Coverage for Chronic Conditions: The high average billing amounts for Cancer and Diabetes—both exceeding USD 25,000—indicate that patients face substantial financial risk. Policymakers and insurance providers should ensure that chronic disease management is comprehensively covered under existing or newly developed insurance products, and should explore subsidised coverage schemes for vulnerable populations.
4. Investigate Clinical Determinants of Billing: Since neither patient age nor admission type significantly predicts billing amount, future research should examine clinical variables such as length of hospital stay, number of procedures, medication prescribed, and test results as potential cost drivers. This would enable more precise financial forecasting and targeted cost-efficiency measures.
5. Apply This Framework to Real Institutional Data: The analytical approach demonstrated in this study should be replicated using actual hospital admission records from local facilities. This would generate contextually valid insights directly applicable to institutional planning, and would support the development of a data-driven culture within the healthcare sector.

8. REFERENCES

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